

Timeline Check In

Macey Hartmann and Erick Morales Oyola

2026-05-18

Ideas for analysis and background

Should include some citations (not all)

Should include ideas for analyses to answer questions and some preliminary code to run those analyses

Visualizations directly relevant to answering questions

Figure 1: Dominant species percent cover across survey years

```
fig1 <- ggplot(data = dominant_by_year,
               aes(x = year,
                   y = mean_pc,
                   fill = psoc_grouped)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c(
    "Salicornia_pacifica"      = "#0D5E45",
    "Distichlis_spicata"      = "#1D9E75",
    "Atriplex_lentiformis"    = "#5DCAA5",
    "Eleocharis_macrostachya" = "#A8DBC9",
    "Raphanus_sativus"       = "#D85A30"
  )) +
  scale_x_continuous(breaks = seq(2018, 2025, 1)) +
  labs(x = "Survey year",
       y = "Mean percent cover (%)",
       fill = "Species") +
  theme_minimal() +
  theme(legend.position = "bottom",
```

```

legend.text = element_text(face = "italic"),
legend.title = element_text(face = "bold"))

```

fig1

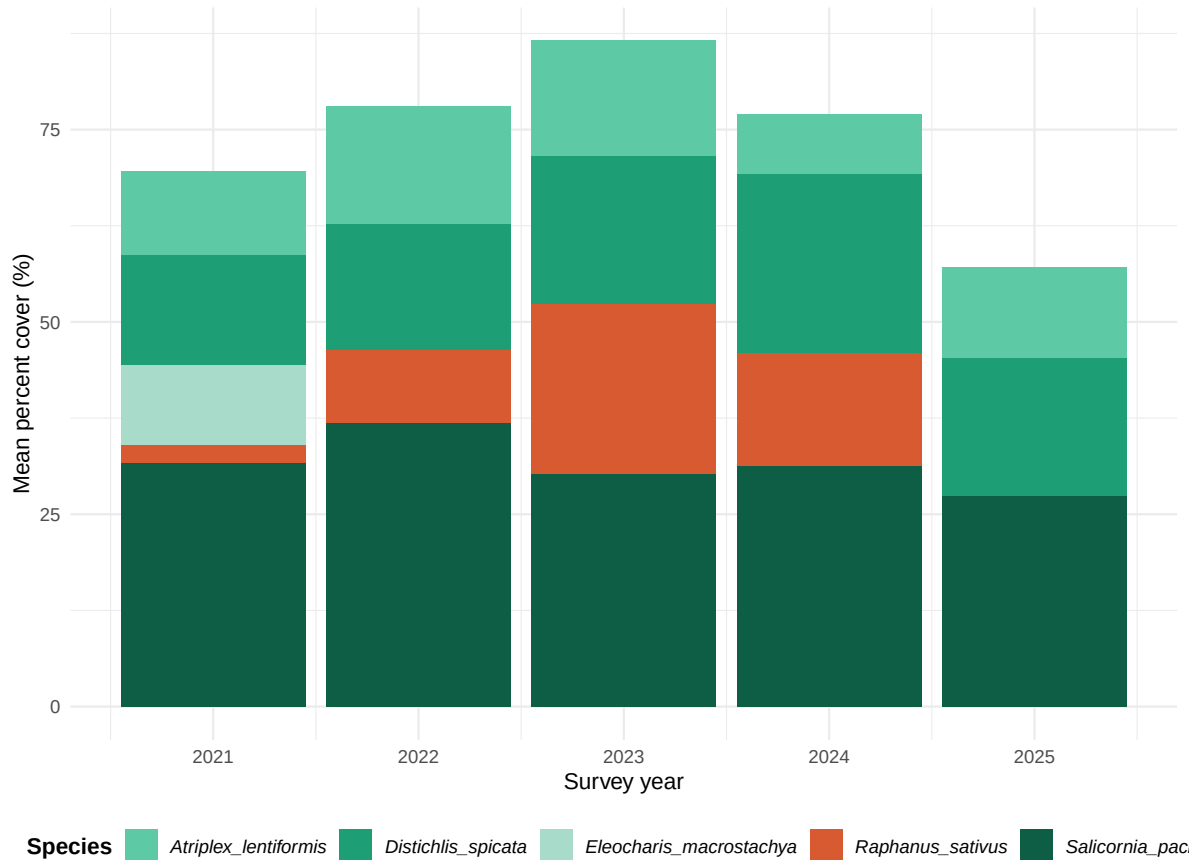


Figure 1: Mean percent cover of the five dominant salt marsh plant species at NCOS across survey years, averaged across transects within each year. Dominant species were identified as the top five by mean percent cover across all survey years. Bar height reflects total mean cover of the five dominant species and varies across years. Green shades indicate native species; coral indicates non-native species.

Figure 2: Proportional native vs. non-native cover across survey years

```

fig2 <- ggplot(data = native_nonnative_raw,
               aes(x = year,

```

```

        y = mean_cover,
        color = cover_category,
        fill = cover_category)) +
geom_ribbon(aes(ymin = mean_cover - se_cover,
               ymax = mean_cover + se_cover),
           alpha = 0.2,
           color = NA) +
geom_line(linewidth = 1) +
geom_point(size = 3) +
scale_color_manual(values = c("NATIVE COVER" = "#1D9E75",
                              "NON-NATIVE COVER" = "#D85A30")) +
scale_fill_manual(values = c("NATIVE COVER" = "#1D9E75",
                              "NON-NATIVE COVER" = "#D85A30")) +
scale_x_continuous(breaks = seq(2018, 2025, 1)) +
labs(x = "Survey year",
     y = "Mean percent cover ( $\pm$  1 SE)",
     color = "Cover category",
     fill = "Cover category") +
theme_minimal() +
theme(legend.position = "bottom",
      legend.title = element_text(face = "bold"))

```

fig2

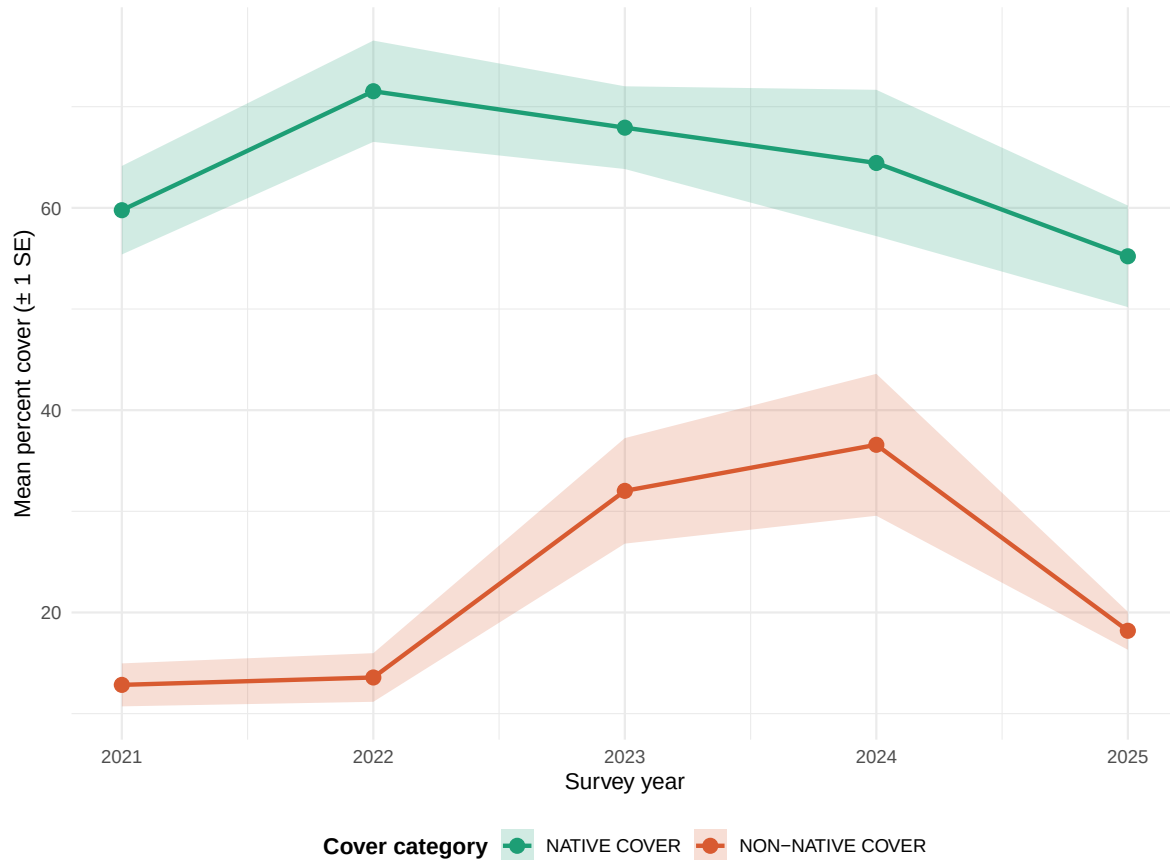


Figure 2: Mean percent cover (± 1 SE) of native and non-native vegetation in salt marsh habitat at NCOS across survey years. Points represent means and shaded bands represent ± 1 standard error. Native cover is shown in green; non-native cover in coral.

Figure 3: Species rank plot

```
# fig3 - pointrange instead of col
cutoff_x <- mean(c(species_rank$mean_pc[5], species_rank$mean_pc[6]))

fig3 <- ggplot(data = species_rank_top20,
  aes(x = mean_pc,
    y = reorder(psoc, mean_pc),
    color = cover_category)) +
  geom_pointrange(aes(xmin = mean_pc - se_pc,
    xmax = mean_pc + se_pc)) +
  geom_vline(xintercept = cutoff_x,
```

```

        linetype = "dashed",
        color = "gray50") +
scale_color_manual(values = c("NATIVE COVER" = "#1D9E75",
                              "NON-NATIVE COVER" = "#D85A30")) +
labs(x = "Mean percent cover ( $\pm$  1 SE)",
     y = "Species",
     color = "Cover category") +
theme_minimal() +
theme(axis.text.y = element_text(face = "italic", size = 9),
      legend.position = "bottom",
      legend.title = element_text(face = "bold"))

```

fig3

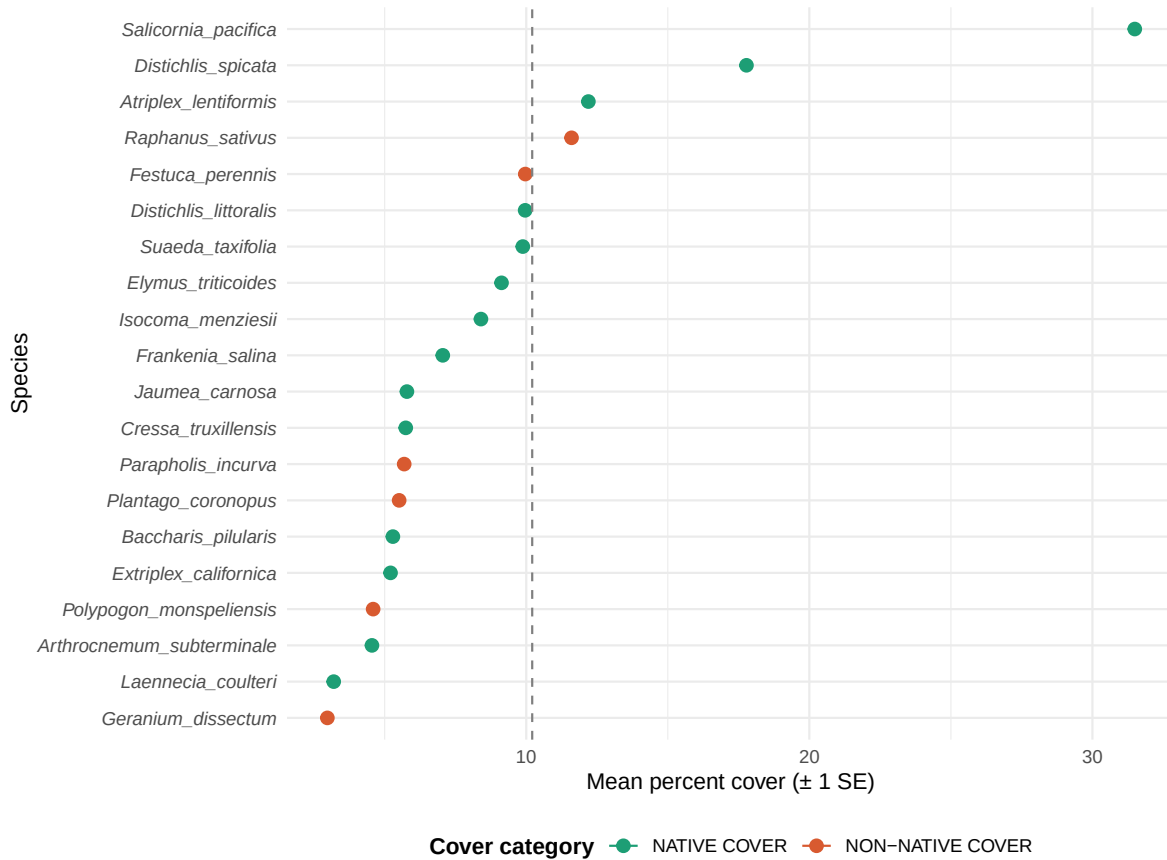


Figure 3: Mean percent cover (± 1 SE) of the top 20 plant species recorded in salt marsh habitat at NCOS, averaged across transects and survey years and ranked from highest to lowest mean cover. Points represent means and horizontal bars represent ± 1 standard error. Green indicates native species; coral indicates non-native species. Dashed line marks the cutoff for the five dominant species visualized in figure 1.

Other exploratory visualizations

Figure 4: Total vegetation cover across survey years

```
fig4 <- ggplot(data = total_cover,
               aes(x = year,
                   y = mean_total)) +
  geom_ribbon(aes(ymin = mean_total - se_total,
                  ymax = mean_total + se_total),
```

```

    fill = "#1D9E75",
    alpha = 0.2) +
geom_line(color = "#1D9E75",
    linewidth = 1) +
geom_point(color = "#1D9E75",
    size = 3) +
labs(x = "Survey year",
    y = "Total mean percent cover (%)") +
theme_minimal()

```

fig4



Figure 4: Total mean percent vegetation cover (native + non-native combined) in salt marsh habitat at NCOS across survey years. Shaded band represents ± 1 standard error around the mean.

Plan for elective (if completing)

Should include some description/image of drafts

Should include plan for making progress in the next two weeks